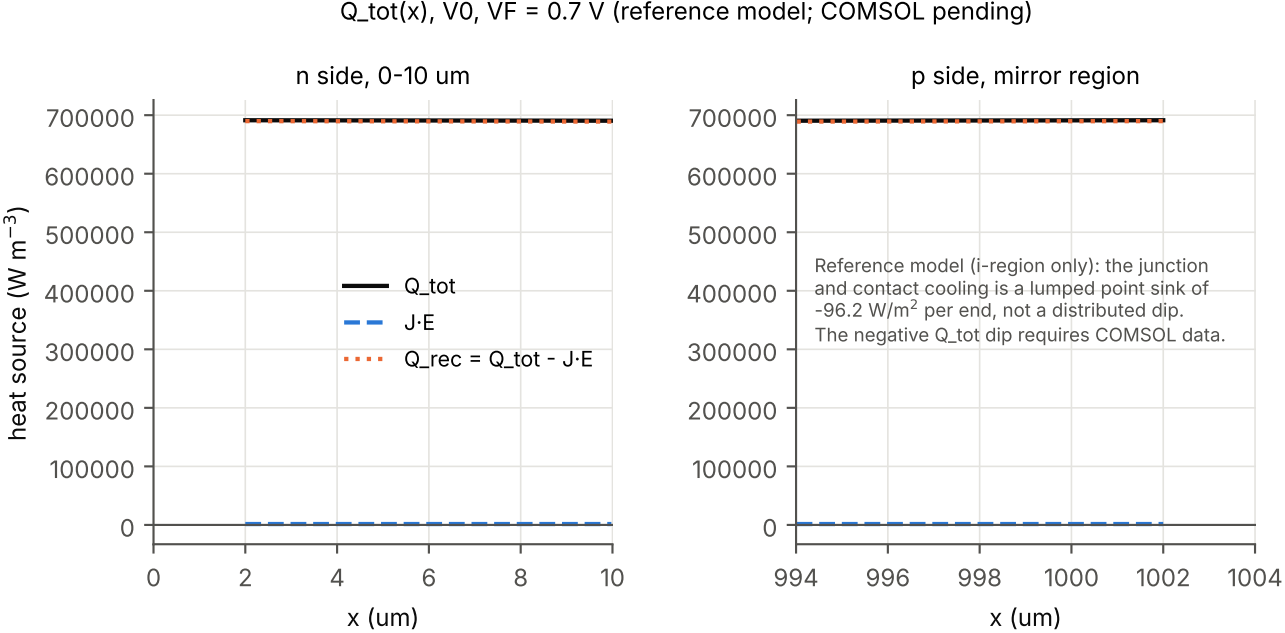
Q_tot(x), V0, VF = 0.7 V (reference model; COMSOL pending)
n side, 0-10 um
p side, mirror region
heat source (W m^{-3})
x (um)
x (um)
Q_tot
J·E
Q_rec = Q_tot - J·E
Reference model (i-region only): the junction and contact cooling is a lumped point sink of -96.2 W/m^2 per end, not a distributed dip.
The negative Q_tot dip requires COMSOL data.
700000
600000
500000
400000
300000
200000
100000
0
0
2
4
6
8
10
994
996
998
1000
1002
1004